GRIDGUARD AI 4.0
National Smart Grid Executive Intelligence Report

Generated: 2026-02-09T04:33:45.848Z
Scope: National operational risk, theft intelligence, transformer reliability, alert posture, and recommended actions.
Source: Live platform data (regions, alerts, transformer telemetry, meter anomalies, and reading streams).

Executive Narrative

This executive brief presents a formal live operational assessment of GRIDGUARD AI 4.0 based on current telemetry, anomaly outputs, and alert intelligence.

At the time of generation, national average theft score stands at 22.14% and national distribution loss stands at 7.45%. The transformer risk index is 41.86% and the carbon optimization score is 70.43%.

System footprint currently includes 35 regions, 120 transformers, and 1080 smart meters.

Risk zoning currently indicates 0 red zones, 1 yellow zones, and 34 green zones.

The estimated daily financial impact of loss is INR 28,801.19 based on recent load behavior and live loss conditions.

GRIDGUARD AI Executive Summary (National Operations)

* **System Stability:** Regional

Key Metrics

National Theft Score (Avg): 22.14%
National Distribution Loss (Avg): 7.45%
Transformer Risk (Avg): 41.86%
Carbon Optimization (Avg): 70.43%
Open Alerts: 127
High Alerts (24h): 5
Estimated Daily Financial Impact (INR): 28,801.19

National Operational Snapshot

Total Regions Monitored: 35
Total Transformers Monitored: 120
Total Smart Meters Monitored: 1080
National Theft Score (Avg): 22.14%
National Distribution Loss (Avg): 7.45%
National Transformer Risk (Avg): 41.86%
National Carbon Optimization Score (Avg): 70.43%

Risk Zone Distribution

Red Zones (>70 theft score): 0

Yellow Zones (40-70 theft score): 1
Green Zones (<40 theft score): 34

Alert Intelligence

Total Alerts Logged: 135
Open Alerts (Unacknowledged): 127
High Severity Alerts in Last 24h: 5

- [2026-02-09T04:33:36.474Z] (MEDIUM) Gujarat: Anomaly detected for meter 4066bb65-0186-431b-88e1-7f38360aefa7
- [2026-02-09T04:32:39.062Z] (MEDIUM) Tamil Nadu: Anomaly detected for meter 402b9994-7208-47d7-bedf-4042d0970602
- [2026-02-09T04:32:08.843Z] (HIGH) Chandigarh: Anomaly detected for meter f67523ab-91ba-421a-804b-1401ab2cf4e7
- [2026-02-09T04:32:05.801Z] (HIGH) Gujarat: Anomaly detected for meter ffbc2b71-4b05-4b90-9f22-6c0db7faf23a
- [2026-02-09T04:28:32.312Z] (MEDIUM) Delhi: Anomaly detected for meter 213bce00-b18f-4ba1-b67f-9d26e69c56d5
- [2026-02-09T04:28:23.268Z] (MEDIUM) Tamil Nadu: Anomaly detected for meter e3e09d2c-5454-4771-a5b9-4cd3f367387e
- [2026-02-08T06:59:34.922Z] (MEDIUM) Maharashtra: Anomaly detected for meter d1eb1964-df64-46d8-86bd-114a0451928f
- [2026-02-08T06:59:04.759Z] (MEDIUM) Delhi: Anomaly detected for meter 8cca7c5e-4b43-4406-a410-59eb8d6afb92
- [2026-02-08T06:58:52.689Z] (MEDIUM) Delhi: Anomaly detected for meter 582a0594-17eb-4f71-95ad-95c602685831
- [2026-02-08T06:58:43.638Z] (MEDIUM) Karnataka: Anomaly detected for meter d5fa83ba-cc64-4320-83d9-ff2810edae74
- [2026-02-08T06:58:40.623Z] (MEDIUM) Bihar: Anomaly detected for meter 94ca19e9-46f6-4855-b709-df8df4b4a861
- [2026-02-08T06:58:31.572Z] (HIGH) Maharashtra: Anomaly detected for meter 23849cd3-abef-4eee-a2c7-e89b2f3c2069

Top High-Risk Regions

- 1. Gujarat | Theft 58% | Loss 19.47% | Transformer Risk 71.29% | Carbon 70% | Updated 2026-02-09T04:33:36.474Z
- 2. Mizoram | Theft 34% | Loss 11.29% | Transformer Risk 43.27% | Carbon 59% | Updated 2026-02-07T22:17:27.901Z
- 3. West Bengal | Theft 34% | Loss 6.71% | Transformer Risk 41.29% | Carbon 73% | Updated 2026-02-07T22:17:27.901Z
- 4. Sikkim | Theft 33% | Loss 5.17% | Transformer Risk 48.68% | Carbon 85% | Updated 2026-02-07T22:17:27.901Z
- 5. Himachal Pradesh | Theft 32% | Loss 9.67% | Transformer Risk 27.63% | Carbon 64% | Updated 2026-02-07T22:17:27.901Z
- 6. Dadra and Nagar Haveli | Theft 30% | Loss 4.91% | Transformer Risk 23.51% | Carbon 62% | Updated 2026-02-07T22:17:27.901Z
- 7. Arunachal Pradesh | Theft 29% | Loss 9.62% | Transformer Risk 73.5% | Carbon 70% | Updated 2026-02-09T04:33:09.263Z
- 8. Jharkhand | Theft 29% | Loss 6.51% | Transformer Risk 45.79% | Carbon 67% | Updated 2026-02-07T22:17:27.901Z

- 9. Andhra Pradesh | Theft 27% | Loss 8.91% | Transformer Risk 72.84% | Carbon 70% | Updated 2026-02-09T04:30:57.029Z
- 10. Bihar | Theft 26% | Loss 8.78% | Transformer Risk 47.48% | Carbon 70% | Updated 2026-02-09T04:33:39.489Z

Top Vulnerable Transformers

- 1. Delhi | Transformer d232ecfb | Load 74.48% | Temp 54.1 C | Health 0.38 | Risk GREEN
- 2. Arunachal Pradesh | Transformer 3214d04c | Load 73.5% | Temp 63.18 C | Health 0.39 | Risk GREEN
- 3. Andhra Pradesh | Transformer 9ab80d8c | Load 72.84% | Temp 68.3 C | Health 0.39 | Risk GREEN
- 4. Tamil Nadu | Transformer df0db96a | Load 72.09% | Temp 50.65 C | Health 0.4 | Risk GREEN
- 5. Delhi | Transformer 67102a47 | Load 71.89% | Temp 56.63 C | Health 0.4 | Risk GREEN
- 6. Bihar | Transformer bb9efdb5 | Load 71.68% | Temp 47.72 C | Health 0.4 | Risk GREEN
- 7. Gujarat | Transformer 52a08580 | Load 71.29% | Temp 61.76 C | Health 0.41 | Risk YELLOW
- 8. Delhi | Transformer 51302981 | Load 71.29% | Temp 49.83 C | Health 0.41 | Risk GREEN
- 9. Chandigarh | Transformer 1992130b | Load 70.39% | Temp 50.57 C | Health 0.41 | Risk GREEN
- 10. Tamil Nadu | Transformer 8f9fdda4 | Load 69.92% | Temp 56.22 C | Health 0.42 | Risk GREEN

Top Meter Anomalies (7-day window)

- 1. Arunachal Pradesh | Meter 13821bee-947 | Theft Probability 84% | Class RED | Detected 2026-02-07T23:20:42.183Z
- 2. Gujarat | Meter 81ba998c-1b0 | Theft Probability 83% | Class RED | Detected 2026-02-07T23:26:38.082Z
- 3. Delhi | Meter 9657afa8-a5b | Theft Probability 83% | Class RED | Detected 2026-02-07T21:50:38.095Z
- 4. Delhi | Meter 70214083-d40 | Theft Probability 82% | Class RED | Detected 2026-02-07T23:43:33.695Z
- 5. Gujarat | Meter 375a665d-c07 | Theft Probability 82% | Class RED | Detected 2026-02-07T22:36:28.438Z
- 6. Karnataka | Meter 7ee61712-207 | Theft Probability 81% | Class RED | Detected 2026-02-08T06:58:07.420Z
- 7. Karnataka | Meter 417f1359-105 | Theft Probability 81% | Class RED | Detected 2026-02-07T22:42:26.999Z
- 8. Tamil Nadu | Meter 98223e50-2a8 | Theft Probability 81% | Class RED | Detected 2026-02-07T22:36:04.297Z
- 9. Maharashtra | Meter d8ef16cf-5fa | Theft Probability 80% | Class RED | Detected 2026-02-07T22:55:52.712Z
- 10. Tamil Nadu | Meter 74331734-0ba | Theft Probability 79% | Class RED | Detected 2026-02-07T23:38:07.588Z

24-Hour Load Trend

Average Power (24h): 2.131 kW
Estimated Daily Energy Loss: 4114.46 kWh

- 2026-02-08T06:30:00.000Z | Avg Power 2.185 kW | Readings 298
- 2026-02-09T03:30:00.000Z | Avg Power 2.13 kW | Readings 28
- 2026-02-09T04:30:00.000Z | Avg Power 2.078 kW | Readings 52

Recommended Actions

- Maintain active anomaly surveillance and sustain monthly anti-theft audit cadence in medium-risk states.
- Current national loss is within an acceptable band; continue preventive maintenance and loss monitoring.
- Initiate a 14-day transformer reliability sprint focused on high-load assets with low health index.
- Establish an alert triage war-room with SLA-based acknowledgement and closure tracking.
- Use LLM assistant for weekly executive briefings, including state-wise risk rationale and recommended field actions.

Methodology Note

This report is generated from live PostgreSQL operational data and near-real-time simulator/ingest streams.
Risk zone classification follows GRIDGUARD thresholds based on theft probability and transformer stress signals.
Financial impact estimation uses current average power, meter coverage, and loss percentage with an indicative INR conversion factor.

Generated by GRIDGUARD AI 4.0 automated reporting engine.